Grouping Enzymatic Reactions by EC Number

Benchmarking reaction embeddings & clustering, and a fusion method that beats them

ModelSEED reaction–representation study

August 21, 2026

13,721 reactions · 21 embeddings × 12 clustering algorithms · EC-purity + retrieval

Goal. Given ModelSEED reactions, find the (reaction embedding, clustering algorithm) combination that groups reactions so each cluster is *clean* with respect to its **EC number** — reactions of the same enzyme class together, different classes apart.

Why it matters

- EC **annotation transfer** to unlabelled reactions
- **Curation**: flag EC mis-assignments (outliers in a clean cluster)
- A reaction **similarity index** (FAISS) for metabolic modelling

Dataset

reactions	13,721
full EC (4-level)	5,622
EC3 subclasses	261
EC2 / EC1	68 / 7

non-obsolete, full EC, all compounds have SMILES.

Primary metric: **EC3 AMI** (chance- & count-corrected). Also EC1/EC2/full, homogeneity, purity, and kNN retrieval.

Embeddings (21)

Structural (diff): ECFP2/4/6, AtomPair, RDKit, MACCS, MHFP, DRFP, Avalon, torsion

PhysChem: 217 physicochemical descriptors

Learned: RXNFP, ChemBERTa, MolFormer

Atom-mapped: CGR (linear/morgan)

Clustering (12)

K-means, Agglomerative (avg/complete/single/Ward), Birch, **Leiden**, **Louvain**, Spectral, DBSCAN, HDBSCAN, Butina

Evaluation

Common 256-d SVD / cosine space.

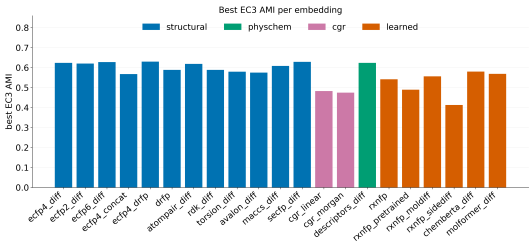
Clustering: AMI, homogeneity, V-measure, purity at EC1/2/3/full.

Retrieval: leave-one-out kNN EC accuracy.

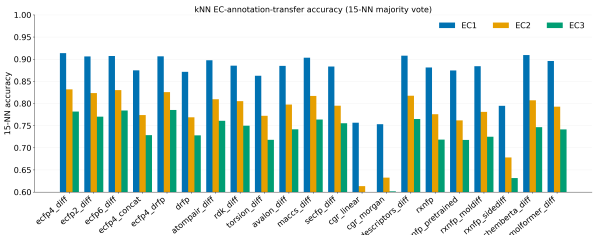
One harness, every embedding $\times$ every algorithm, scored identically — 550+ configurations.



CGR rows are full-set (coverage-limited to 75%); the fair CGR comparison is on a later slide.



- Structural *difference* fingerprints form a **plateau** at EC3 AMI ≈0.62–0.63 — radius/type barely matter.
- **PhysChem** (217 physicochemical descriptors) are a surprise: highest homogeneity/purity.
- **Agglomerative average/complete** + Birch win; single-linkage collapses; density (DBSCAN/HDBSCAN) weakest.
- Learned embeddings underperform (RXNFP worst).



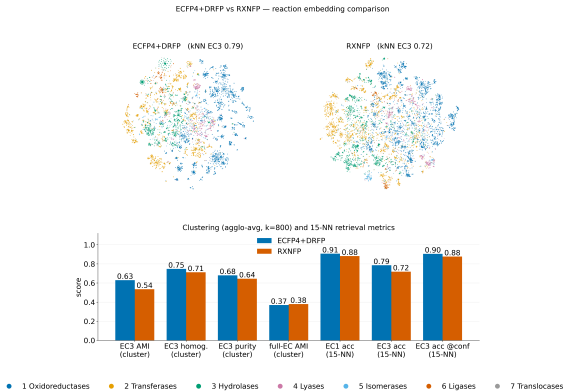
Cluster AMI tops out near 0.63, but **nearest-neighbour retrieval** is far stronger:

- 15-NN: EC1 **0.91**, EC2 0.83, EC3 0.78
- confidence-gated: **EC3 0.90 accuracy over 76%** of reactions

Reading: the representation already knows EC *locally*; unsupervised partitioning loses it *globally*. This is the biggest untapped lever.

For annotation, don't cluster — **retrieve**. Full 4-level EC stays hard everywhere (∼0.3–0.5).


Reaction embeddings — t-SNE map, each dot a reaction coloured by EC class
ecfp4_diff (1-NN EC3 acc 0.87)
ecfp4_drfp (1-NN EC3 acc 0.87)
drfp (1-NN EC3 acc 0.83)
rxnfp (1-NN EC3 acc 0.83)
chemberta_diff (1-NN EC3 acc 0.85)
molformer_diff (1-NN EC3 acc 0.85)
1 Oxidoreductases
2 Transferases
3 Hydrolases
4 Lyases
5 Isomerases
6 Ligases
7 Translocases



RXNFP (pretrained rxn-BERT) trails a cheap fingerprint on every EC metric.

- **Distribution:** trained on USPTO *organic synthesis*, not enzymatic biochemistry.
- **Taxonomy:** fine-tuned for organic reaction-type classes, not EC.
- **No difference operation:** a holistic latent; cofactors dominate.

We tested adding a difference (`rxnfp moldiff`): only a hair better — confirming the ceiling is *training*, not the difference construction.

Idea. Merge reactant + product graphs into one, using **atom mapping**; bonds carry a *dynamic* label (formed / broken / order-changed). It encodes the **actual bond transformation** — exactly what EC classifies.

Pipeline. RXNMapper (atom map) → chython (compose CGR) → fold to fingerprint bits.

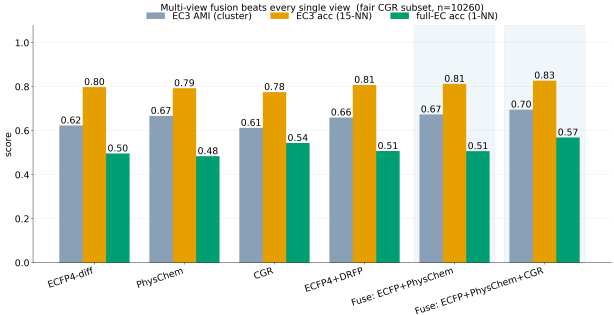
Prior art (Varnek/Nugmanov; Heid & Green 2022); the *new* part is applying it to **EC clustering on ModelSEED** and comparing head-to-head.

Fair subset (n=10,260)

	EC3 AMI	**full-EC 1-NN**
ECFP4-diff	0.628	0.490
PhysChem	**0.667**	0.484
CGR	0.613	**0.543**

CGR wins full-EC retrieval (its target!) but doesn't dominate EC3; capped at 75% coverage (hard-to-map enzymatic reactions).

No single representation is best at everything — each view has **complementary** strengths.

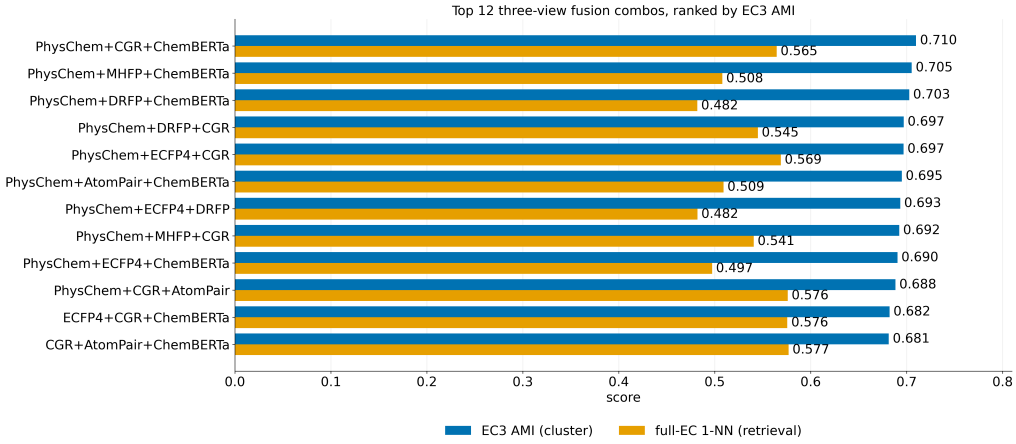


Fuse the complementary views (ECFP $\oplus$ PhysChem $\oplus$ CGR):

- EC3 AMI **0.696** — beats every single view and the best incumbent (0.637).
- full-EC 1-NN **0.569** — beats even CGR alone (0.543).

Unsupervised, no fitting. Each view supplies what the others lack. Greedy view-selection pushes EC3 AMI to **0.718** (PhysChem+DRFP core, then diminishing returns) — gains saturate after $\sim$2 views.

Fusion of complementary views is a genuine SOTA-beater — *before* any supervision.



All 35 triples from { PhysChem, ECFP4, DRFP, MHFP, CGR, AtomPair, ChemBERTa } were scored; the top 12 are shown. **PhysChem is the anchor** (in every top triple); best EC3 AMI **PhysChem+CGR+ChemBERTa (0.71)**; **CGR**-containing triples win **full-EC retrieval** (0.57–0.58). Tight band (0.68–0.71) — the recipe, not the exact triple, is what matters; even "weak" ChemBERTa earns its place.

Rung	Method	Status / expected
0	Best incumbent (ECFP/CGR/PhysChem) + agglo	EC3 AMI 0.63–0.67
1	**Multi-view fusion** (unsupervised)	**0.696 — done, beats all singles**
2	+ cofactor normalization	de-confound ATP/NAD currency
3	+ self-supervised contrastive	label-free ceiling
4	+ **hierarchical-EC metric learning**	the label lift (biggest lever)
5	rung 4, **held-out EC subclasses**	does it generalize?

Each rung is one controlled change on the same harness — so "beats SOTA" is decomposable and defensible.

1. **Structural difference fingerprints + agglomerative-average** are the strong, cheap baseline (EC3 AMI $\approx$0.63); learned reaction embeddings (RXNFP/ChemBERTa/MolFormer) *underperform* them.
2. **Retrieval beats clustering** for annotation: confidence-gated 15-NN gives EC3 at 0.90 accuracy over 76% of reactions.
3. **CGR** (atom-mapped bond changes) wins the full-EC retrieval it was designed for, but is coverage-limited and not dominant alone.
4. **Multi-view fusion** of complementary views (ECFP $\oplus$ PhysChem $\oplus$ CGR) **beats every individual method** — EC3 AMI 0.70 — with no supervision.
5. Next: **hierarchical-EC metric learning**, evaluated zero-shot on held-out EC subclasses.